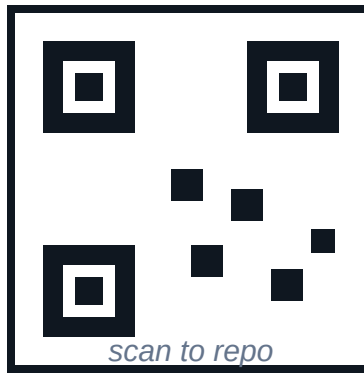


Evaluating an Offline Detector Against Adversarial Prompt Attacks



A Reproducible Framework for Prompt-Attack Generation and Defensive Evaluation

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Security Engineering • AI / Agent Security • Cloud Security

Apache-2.0 • Python 3.12 • HEAD d3fe468 • verified 2026-09-26

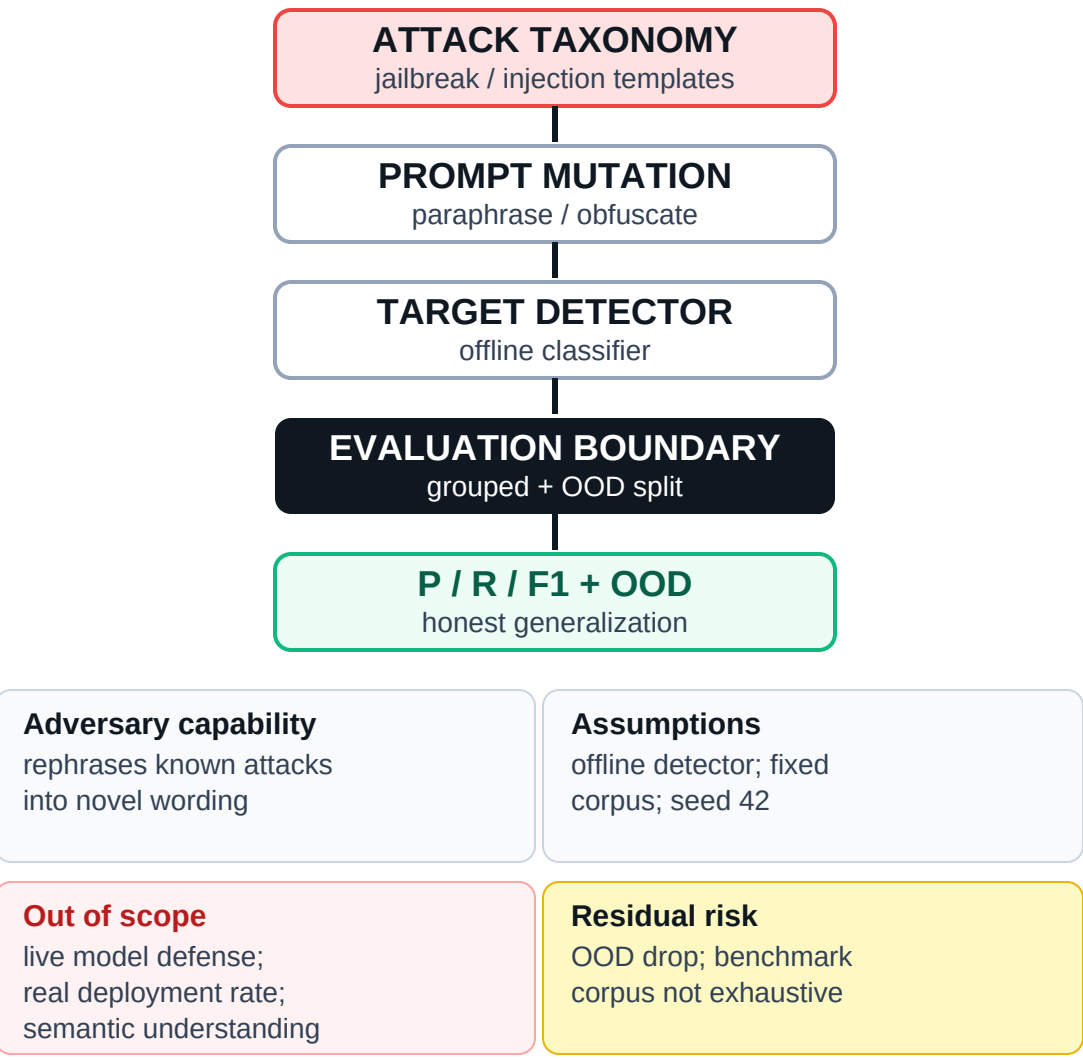
RESEARCH QUESTION

Does a prompt-attack detector generalize beyond the templates it was tuned on — and how much of its headline score is memorized surface structure?

01 • THE FAILURE MODE

A detector benchmarked only on its own attack templates can look near-perfect while failing on rephrased, out-of-distribution attacks. Reporting the in-distribution number alone overstates real defense. This framework generates attacks and evaluates a detector with an explicit OOD generalization split.

02 • THREAT MODEL



03 • OBJECTIVE

Determine how much a prompt-attack detector's score is template memorization by measuring a grouped in-distribution vs OOD paraphrase split.

- 01 – Attack taxonomy + mutation
- 02 – Grouped template split (seed 42)
- 03 – OOD novel-phrasing benchmark
- 04 – External InjectionBench/JailbreakBench

HERO • ATTACK-GEN + EVAL PIPELINE

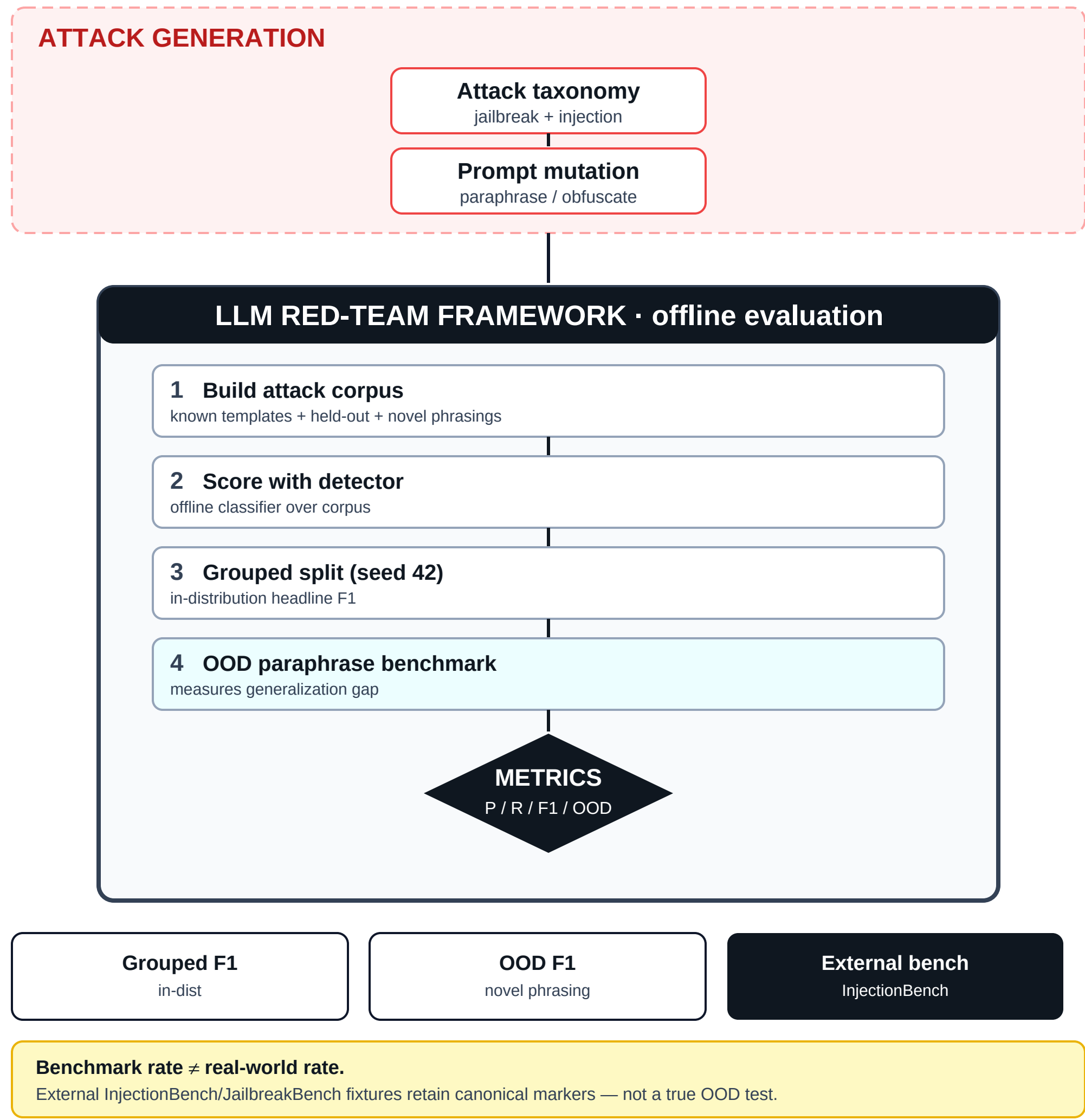


Figure 1. Attacks are generated, mutated, and scored under a grouped split and an OOD paraphrase split that exposes memorization.

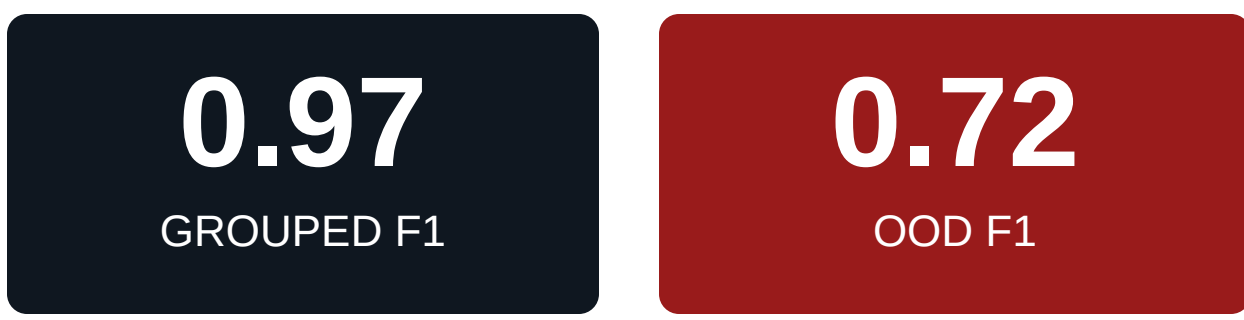
METHOD • EVALUATION SPLITS



CONTRIBUTION

A reproducible prompt-attack generation + evaluation framework that reports both the in-distribution headline (grouped F1 0.97) and the OOD paraphrase result (F1 0.72), quantifying the ~25-point drop attributable to memorized surface structure.

RESULTS • GENERALIZATION GAP (measured)



Grouped template split, seed 42 (test_eval.py, pinned).
OOD novel-phrasing regression — ~25-pt drop = memorization.

Verified metrics	
Random split F1	1.00
OOD recall	0.92
OOD precision	0.59

F1 ACROSS EVALUATION CONDITIONS



OOD is the honest test; the ~25-pt drop from grouped is memorized surface structure. scan.

BEHAVIOR-SCREENING CAVEAT

JailbreakBench behavior screening F1 = 0.15 (harmful-vs-benign goals, NOT injection F1). Labeled explicitly to avoid conflating tasks.

NOT ESTABLISHED BY THIS REPO

Real-world jailbreak detection rate. Live-model defense. That external fixtures are a true OOD test.

SECURITY BOUNDARY

PROTECTS / DETECTS	DOES NOT ESTABLISH
- known attack templates - mutation variants - injection markers - reproducible eval - OOD measurement	- live LLM defense - novel semantic attacks - deployment rate - behavior policy

IN-DIST ≠ OUT-OF-DIST

LIMITATIONS

- 1. OOD F1 0.72 — much of headline is memorized.
- 2. External fixtures keep canonical markers (not true OOD).
- 3. Offline detector; no live-model evaluation.
- 4. Corpus is finite; not exhaustive of real attacks.
- 5. Behavior-screening F1 (0.15) is a different task.

FUTURE WORK

- True held-out semantic OOD corpus.
- Live-model guardrail evaluation.
- Larger, diverse attack taxonomy.
- Human-adversary red-team comparison.
- Calibrated decision thresholds.

CONCLUSION

The framework generates prompt attacks and evaluates a detector under grouped and OOD splits. It reports the full picture: 0.97 in-distribution vs 0.72 OOD, exposing memorization rather than hiding it. It measures a detector; it is not a live defense.

REPRODUCE
pytest tests/
python benchmarks/ood_novel_phrasings.py